

Replication Plan

Variational Monte Carlo Calculations of $A \leq 4$ Nuclei with an Artificial Neural-Network
Correlator Ansatz
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Auto-generated Replication Blueprint

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Paper Overview

This paper applies variational Monte Carlo (VMC) with a neural-network-based wave function ansatz to compute ground-state energies of light nuclei ($A \leq 4$: deuteron, triton, helium-3, helium-4). The neural network serves as a correlator that modifies a Slater-determinant or Jastrow baseline, parameterizing many-body correlations. The network is optimized using the stochastic reconfiguration (SR) algorithm to minimize the variational energy. The nuclear Hamiltonian uses realistic nucleon–nucleon (NN) potentials (e.g., AV6', AV8', or Minnesota potential).

Detailed Workflow

Phase 1: Nuclear Hamiltonian

1.1 Implement the NN potential.

- Central, spin–spin, tensor, spin–orbit components.
- Potential parameters from published tables (Argonne AV6'/AV8' or Minnesota).

1.2 Kinetic energy operator.

- Non-relativistic kinetic energy $T = -\frac{\hbar^2}{2m_N} \sum_i \nabla_i^2$.
- Remove center-of-mass kinetic energy.

Phase 2: Wave Function Ansatz

2.1 Construct the baseline wave function.

- Slater determinant of single-particle orbitals (Gaussians or harmonic oscillator).
- Jastrow correlation factor for short-range pair correlations.

2.2 Neural network correlator.

- Input: inter-particle distances r_{ij} and spin/isospin quantum numbers.
- Architecture: fully connected network with 2–4 hidden layers; tanh or ReLU activations.
- Output: multiplicative correction to the wave function $\Psi = \text{NN}(\{r_{ij}\}) \times \Psi_{\text{baseline}}$.
- Ensure antisymmetry under particle exchange (via the Slater determinant).

Phase 3: VMC Sampling

3.1 Metropolis–Hastings sampling.

- Sample nucleon configurations $\mathbf{R} = \{r_1, \dots, r_A\}$ from $|\Psi(\mathbf{R})|^2$.
- Propose moves: single-particle displacement; tune acceptance rate $\sim 50\%$.

3.2 Compute local energy.

$$E_L(\mathbf{R}) = \frac{H\Psi(\mathbf{R})}{\Psi(\mathbf{R})}$$

3.3 Estimate the variational energy $E_V = \langle E_L \rangle$ and its statistical error.

Phase 4: Stochastic Reconfiguration Optimization

4.1 Compute the overlap matrix S and energy gradient f :

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle$$
$$f_i = \langle E_L O_i \rangle - \langle E_L \rangle \langle O_i \rangle$$

where $O_i = \partial \ln \Psi / \partial \theta_i$.

4.2 Update parameters: $\delta \theta = -\eta S^{-1} f$ (with regularization).

4.3 Iterate until energy converges (typically 10^3 – 10^4 iterations).

Phase 5: Results and Validation

5.1 Compute ground-state energies for ^2H , ^3H , ^3He , ^4He .

5.2 Compare to exact diagonalization (for small systems) and published VMC/GFMC results.

5.3 Reproduce convergence curves (energy vs. iteration) from the paper.

5.4 Study dependence on network architecture (depth, width).

Datasets

Dataset / Source	Type	Location / Notes
NN potential parameters	Published tables	Argonne AV6'/AV8': Wiringa et al. (1995, 2002); Minnesota: Thompson et al. (1977)
Reference energies	Published benchmarks	GFMC results from Carlson et al., Pudliner et al.
All other data	Synthetic (MC samples)	Generated by the VMC code

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
PyTorch	≥ 1.10	Neural network, automatic differentiation https://pytorch.org/
JAX	latest	Alternative: JIT-compiled autodiff https://github.com/google/jax
NumPy	≥ 1.21	Array operations
SciPy	≥ 1.7	Linear algebra (SR matrix inversion)
Matplotlib	≥ 3.4	Energy convergence and wave function plots

Tool / Library	Version	Purpose / URL
NetKet	latest	Optional: VMC framework with SR built in https://www.netket.org/

Hardware Requirements

- $A \leq 4$: Single GPU or multi-core CPU; runs in hours to days.
- **Memory**: Modest (~ 8 GB RAM).